

# Fixed-point existence of self-consistent semiclassical gravity

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Will Regelmann\*   Claude (Anthropic)<sup>†</sup>

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## Abstract

The semiclassical Einstein equation  $G_{\mu\nu}[g] = 8\pi G\langle\hat{T}_{\mu\nu}\rangle_g$  is a fixed-point condition: the geometry and quantum state must be mutually consistent. We address the question of whether such fixed points exist. We present three analyses: (1) exact existence in the cosmological case via the trace anomaly (Starobinsky, 1980) (existence **Rigorous**, by citation; the explicit fixed-point curvature scale  $H_0^2 = 180\pi/(G|a_2|)$  is **Sketch** — a convention-dependent reduction not verbatim in the cited sources, Section 3); (2) a perturbative argument near classical solutions *on globally hyperbolic spacetimes with compact Cauchy surfaces*, via a Banach contraction whose target estimate is a contraction constant  $\kappa \sim (m/M_P)^2 \ll 1$  for massive fields, developed via a Hadamard decomposition of the order-reduced response kernel into a local differential operator and a non-local smooth kernel — this argument is presently a **Sketch**: an earlier Rigorous label was withdrawn after review identified gaps in the kernel bound and in the contraction estimate (Section 4), and the massless case, relevant to photons and gluons, remains open (Section 8); and (3) conditional existence in the general case via the Schauder fixed-point theorem on globally hyperbolic spacetimes with compact Cauchy surfaces, under five stated assumptions of which one remains an open conjecture, plus an explicit semiclassical hypothesis on  $\hbar$  (Section 5). An earlier claim that the Planck scale emerges from these analyses as the natural validity boundary is withdrawn: the derivation chain behind the contraction constant contains a dimensional inconsistency, and neither of its two candidate readings places the breakdown at the Planck scale (Section 7). An effective initial-value formulation, valid order-by-order in  $\kappa$ , is derived from the global fixed-point condition.

*Note:* This is a companion paper to “Gaussian temporal profile of gravitational decoherence from the Einstein-Langevin equation.” The decoherence prediction of that paper does not depend on the existence results derived here; the present paper addresses the independent question of whether the semiclassical framework is internally self-consistent.

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\*Corresponding author.

<sup>†</sup>AI assistant; contributed to mathematical development and analysis.

# 1 Introduction

The semiclassical Einstein equation:

$$G_{\mu\nu}[g] = 8\pi G \langle \Psi | \hat{T}_{\mu\nu}[g] | \Psi \rangle \quad (1)$$

is not a dynamical equation in the usual sense. It is a fixed-point condition: we seek the geometry  $g_{\mu\nu}$  such that the quantum state it hosts, evolving on it, sources exactly that geometry. The position that eq. (1) may be fundamental (rather than an approximation to quantized gravity) dates to Møller (1962) and Rosenfeld (1963) and has been developed by Oppenheim [1]. Whether this position is internally consistent depends on whether fixed points exist.

This paper presents existence arguments at three levels of generality. The cosmological result (Section 3) is exact. The perturbative Banach-contraction argument (Section 4) is labeled **Sketch**: it was promoted to Rigorous in an earlier revision, and that label has been withdrawn after review identified specific gaps, catalogued where they occur. The Schauder argument (Section 5) remains conditional on an open conjecture (A3) and an explicit semiclassical hypothesis on  $\hbar$ . An earlier claim that the Planck scale emerges from the convergence analysis as the natural validity boundary is withdrawn pending a corrected contraction estimate (Section 7).

The perturbative contraction estimate is permanently Sketch at the stated generality: the binding obstruction (Gap M3) is mass-independent — the non-local response kernel is a Lorentzian Green operator with causal-past support, not a spatial elliptic resolvent, and this distinction does not depend on field mass — so the same Gap (M3) obstruction applies equally to massless fields such as the photon and gluon (Section 8). The estimate also requires a compact Cauchy surface, so it does not cover Minkowski space or asymptotically flat backgrounds (Section 8).

## 2 The self-consistency map

Define  $\mathcal{F} : g_{\mu\nu} \mapsto g'_{\mu\nu}$  by:

1. Given trial geometry  $g$ , construct the QFT on  $(\mathcal{M}, g)$  and identify the physical state  $|\Psi_g\rangle$  satisfying the Hadamard condition.
2. Compute  $S_{\mu\nu}[g] \equiv 8\pi G \langle \hat{T}_{\mu\nu} \rangle_{g, \text{ren}}$ .
3. Solve  $G_{\mu\nu}[g'] = S_{\mu\nu}[g]$  for  $g'$ .

We seek  $g^*$  such that  $\mathcal{F}(g^*) = g^*$ .

**State selection.** Step 1 requires care: the Hadamard condition constrains the wavefront set of the two-point function but does not uniquely select a state. In the cosmological case (Section 3), de Sitter symmetry selects the conformal vacuum. In the perturbative case (Section 4), the state on the corrected geometry is the unique perturbation of the state on the classical background. For the Schauder argument (Section 5), multi-valuedness is handled by restricting to a suitable branch or by applying Kakutani’s theorem for set-valued maps.

**Single-valuedness assumption.** The Banach argument (Section 4) assumes  $\mathcal{F}$  is single-valued: in the perturbative regime, the Hadamard state on the corrected geometry is

the unique perturbation of the state on the classical background, so no state-selection ambiguity arises. The Schauder argument (Section 5) likewise assumes single-valuedness via the branch restriction noted above. An extension to the genuinely set-valued case via Kakutani’s fixed-point theorem is possible but is not pursued here.

### 3 Cosmological fixed point: exact solution

**Theorem 1** (Starobinsky, 1980 [2]). *The self-consistency equation on FRW spacetime with conformal matter admits an exact de Sitter solution  $a(t) = a_0 e^{H_0 t}$  with:*

$$H_0^2 = \frac{180\pi}{G|a_2|} \quad (2)$$

where  $a_2$  is the coefficient of the Euler density in the trace anomaly. The de Sitter solution is unstable: the quasi-de Sitter stage is long-lived but decays, and in the inflationary application this instability is the graceful-exit mechanism.

(Existence and instability of the de Sitter fixed point: **Rigorous**, by citation to Starobinsky [2]. Explicit coefficient  $H_0^2 = 180\pi/(G|a_2|)$ : **Sketch** — see Rigor status note below.)

*Proof sketch.* For constant  $H$ , the trace anomaly [3] reduces to  $\langle \hat{T}^\mu{}_\mu \rangle = -a_2 H_0^4/120\pi^2$ . The Friedmann equation yields  $H_0$ . The instability of the de Sitter stage follows from the linearized equation for  $\delta H$  [2].  $\square$

**Rigor status (red-team demotion, June 2026).** The *existence and instability* of the anomaly-driven de Sitter fixed point are **Rigorous, by citation** to Starobinsky [2]. The *explicit coefficient*  $H_0^2 = 180\pi/(G|a_2|)$  is **Sketch**: the algebra is internally consistent (combining  $\langle \hat{T}^\mu{}_\mu \rangle = -a_2 H_0^4/120\pi^2$  with the trace of the semiclassical equation,  $-R = 8\pi G \langle \hat{T}^\mu{}_\mu \rangle$  and  $R = 12H_0^2$  on de Sitter, gives  $H_0^2 = 180\pi/(Ga_2)$ ), but two steps are not closed by the cited sources. (i) Starobinsky parametrizes the consistency condition through the effective action of  $N$  conformal fields (a curvature-scale bound set by the field content), not as the closed form  $180\pi/(G|a_2|)$  with “ $a_2$  = Euler-density coefficient”; the  $180\pi$  is a later reparametrization. (ii) Capper–Duff [3] establishes the *existence* of the conformal trace anomaly, but the de Sitter-evaluated, field-content-summed normalization  $\langle \hat{T}^\mu{}_\mu \rangle = -a_2 H_0^4/120\pi^2$  requires fixing a field content and an anomaly-coefficient convention, neither of which that source supplies. Relative to the standard split  $\langle \hat{T}^\mu{}_\mu \rangle = cC^2 - aE$  with  $C^2 = 0$  and  $E = 24H_0^4$  on de Sitter, the factor  $120\pi^2$  amounts to  $a_2 = 2880\pi^2 a$  — a convention internal to this paper. Finally, the absolute value in  $|a_2|$  silently encodes a positivity condition:  $H_0^2 > 0$  requires  $a_2 > 0$  (which holds for standard conformal matter, where the type-A coefficient  $a > 0$ , but is not free). Issue #133 tracks restoring this to Rigorous (fix the convention, cite a modern anomaly-induced-inflation reference for the explicit coefficient, state the  $a_2 > 0$  condition).

*Remark (correction).* An earlier draft of this paper asserted that the de Sitter solution is a stable attractor with exponentially damped perturbations. This misstates the cited source: in Starobinsky’s model the de Sitter stage is unstable (its decay is what ends inflation), and the existence claim — not stability — is what this section uses. Only existence of the exact fixed point is relied upon below.

This constitutes an existence proof: the de Sitter geometry, used to compute the quantum stress-energy of fields living on it, sources exactly itself.

## 4 Perturbative fixed point: Banach contraction

**Rigor status: Sketch (demoted from Rigorous).** The results of this section — Lemma 1 and Theorem 2 — were promoted to Rigorous in an earlier revision (via the Hadamard-decomposition argument recorded in `explorations/2026-03-02-A1-Hs-bound-derivation`). That label has been withdrawn: review identified four gaps, labeled (M1), (M3), (M4), (M5) below, that the arguments do not close. The proofs are retained as the record of the sketch; each gap is flagged where it occurs. Closing them is follow-up work (Section 8).

Let  $\bar{g}$  be a solution of the classical Einstein equations. Define the quantum-corrected map:

$$\mathcal{F}(g) = \bar{g} + (G_{\text{lin}}^{\bar{g}})^{-1} \left[ 8\pi G \langle \hat{T}_{\mu\nu} \rangle_g^{(\text{quantum})} \right] \quad (3)$$

We first establish a bound on the order-reduced response kernel, then use it to prove the contraction estimate.

**Lemma 1** (Order-reduced response kernel bound — **Sketch**). *Let  $\phi$  be a massive scalar field with mass  $m > 0$  on a globally hyperbolic spacetime  $(\mathcal{M}, \bar{g})$  with compact Cauchy surface  $\Sigma$ . After Parker–Simon order reduction [13], the order-reduced response kernel  $\mathcal{K}_{\mu\nu\alpha\beta}^{\text{red}}(x, x') \equiv \delta \langle \hat{T}_{\mu\nu} \rangle_{\text{ren}}^{\text{red}} / \delta g^{\alpha\beta}(x')$  satisfies*

$$\|\mathcal{K}^{\text{red}}\|_{H^s \rightarrow H^{s-2}} \leq C_K \cdot \frac{\hbar m^2}{(4\pi)^2} \quad (4)$$

uniformly for all  $g \in K_\rho$ , where  $C_K$  is a dimensionless constant depending on  $\rho$ ,  $s$ , and the background geometry  $\bar{g}$ .

*Proof.* We decompose the response kernel via the Hadamard parametrix into a local differential operator and a non-local smooth kernel, bound each separately, and combine via the triangle inequality.

*Step 1: Hadamard decomposition of the response kernel.* By assumption (A1), the renormalized stress-energy tensor is obtained by point-splitting with Hadamard subtraction [9]:

$$\langle \hat{T}_{\mu\nu}(x) \rangle_{\text{ren}} = \lim_{x' \rightarrow x} \mathcal{D}_{\mu\nu}(x, x') W(x, x') + C_{\mu\nu}[g](x) \quad (5)$$

where  $W = W_2 - H$  is the smooth Hadamard remainder ( $W_2$  the full two-point function,  $H$  the universal Hadamard singular part determined by local geometry via the Hadamard recursion),  $\mathcal{D}_{\mu\nu}$  is the second-order point-splitting differential operator, and  $C_{\mu\nu}[g]$  collects the finite local curvature counterterms (involving  $R$ ,  $R_{\mu\alpha\nu\beta}R^{\alpha\beta}$ ,  $\square R$ ).

The functional derivative  $\delta / \delta g^{\alpha\beta}(y)$  of eq. (5) yields three types of contributions to the response kernel  $\mathcal{K}_{\mu\nu\alpha\beta}(x, y)$ :

- (I) Varying the smooth remainder  $W$  — non-local, mediated by the massive field propagator;
- (II) Varying the point-splitting operator  $\mathcal{D}_{\mu\nu}$  — local, from Christoffel symbol variation at  $y = x$ ;
- (III) Varying the curvature counterterms  $C_{\mu\nu}$  — local.

*Step 2: Order reduction and local/non-local decomposition.* Before order reduction, Term (III) involves up to four derivatives of the metric perturbation (from the  $\square R$  counterterm — the source of the Horowitz runaway instability [4]). After Parker–Simon order reduction [13] (assumption (A5)), the fourth-derivative terms are perturbatively eliminated using the classical Einstein equation, leaving Term (III) as a differential operator of order  $\leq 2$ .

Term (II) is a differential operator of order  $\leq 1$ : varying covariant derivatives introduces at most one derivative of  $\delta g$  via the Christoffel symbols. Term (I) decomposes into contact contributions at  $y = x$  (absorbed into the local part) and a smooth non-local kernel for  $y \neq x$ .

Define  $\mathcal{K}_{\text{loc}}^{\text{red}}$  as the sum of all local (distributional) contributions: a differential operator of order  $\leq 2$  on  $\Sigma$ . Define  $\mathcal{K}_{\text{nl}}$  as the remaining non-local smooth kernel.

*Step 3: Bound on the local part.* The local part takes the form  $\mathcal{K}_{\text{loc}}^{\text{red}} = \sum_{|\gamma| \leq 2} a_\gamma(x) \partial^\gamma$  with smooth coefficients whose  $L^\infty(\Sigma)$  norms are set by the one-loop scale:  $\|a_\gamma\|_{L^\infty} \leq C_{\text{loc}} \cdot \hbar m^2 / (4\pi)^2$ . A differential operator of order  $d$  with  $L^\infty$  coefficients maps  $H^s(\Sigma) \rightarrow H^{s-d}(\Sigma)$  boundedly (by the Leibniz rule and Sobolev embedding on compact manifolds). Therefore:

$$\|\mathcal{K}_{\text{loc}}^{\text{red}}\|_{H^s \rightarrow H^{s-2}} \leq C_{\text{loc}} \cdot \frac{\hbar m^2}{(4\pi)^2}. \quad (6)$$

*Step 4: Bound on the non-local part.* For a massive field ( $m > 0$ ), the smooth remainder  $W$  depends on the metric through the Klein–Gordon equation  $(\square_g + m^2 + \xi R)\phi = 0$ . Its functional derivative  $\delta W / \delta g^{\alpha\beta}(y)$  is mediated by the massive Klein–Gordon resolvent on  $\Sigma$ , whose integral kernel decays exponentially:  $|(-\Delta_\Sigma + m^2 + \xi R|_\Sigma)^{-1}(x, y)| \leq C e^{-m \cdot d_\Sigma(x, y)}$  for  $d_\Sigma(x, y) \gg 1/m$ . This is a standard property of elliptic resolvents with positive mass on compact Riemannian manifolds.

Consequently,  $\mathcal{K}_{\text{nl}}$  has a smooth kernel on  $\Sigma \times \Sigma$  with exponential decay. A smooth integral operator on a compact manifold is a smoothing operator, bounded  $H^s \rightarrow H^{s'}$  for all  $s, s'$ ; its  $H^s \rightarrow H^{s-2}$  operator norm is bounded by its Hilbert–Schmidt norm:

$$\|\mathcal{K}_{\text{nl}}\|_{H^s \rightarrow H^{s-2}} \leq \|K_{\text{nl}}\|_{L^2(\Sigma \times \Sigma)} \leq C_{\text{nl}} \cdot \frac{\hbar m^2}{(4\pi)^2}. \quad (7)$$

*Step 5: Combined bound.* By the triangle inequality,  $\|\mathcal{K}^{\text{red}}\|_{H^s \rightarrow H^{s-2}} \leq \|\mathcal{K}_{\text{loc}}^{\text{red}}\| + \|\mathcal{K}_{\text{nl}}\| \leq C_{\mathcal{K}} \cdot \hbar m^2 / (4\pi)^2$ , establishing (4).

*Step 6: Uniformity over  $K_\rho$ .* By assumption (A2), the map  $g \mapsto \mathcal{K}^{\text{red}}[g]$  is continuous. The set  $K_\rho$  is compact in  $H^{s'}$  for  $s' < s$  by Rellich–Kondrachov (established in Section 5). A continuous function on a compact set attains its supremum, so  $\sup_{g \in K_\rho} \|\mathcal{K}^{\text{red}}[g]\|_{\text{op}} < \infty$ . The constant  $C_{\mathcal{K}}$  absorbs this supremum.  $\square$

**Gap (M3): hyperbolic-to-elliptic substitution in Step 4.** The variation  $\delta W / \delta g^{\alpha\beta}(y)$  is governed by the *Lorentzian* (hyperbolic) Klein–Gordon equation on the spacetime: varying the field equation gives  $(\square_g + m^2 + \xi R) \delta W_2 = -\delta(\square_g + \xi R) W_2$ , whose solution operator has light-cone support. Step 4 instead bounds the variation using the *spatial elliptic* resolvent  $(-\Delta_\Sigma + m^2 + \xi R|_\Sigma)^{-1}$  and its exponential decay. For massive Lorentzian propagators, exponential decay holds only at *spacelike* separation; a metric perturbation at  $y$  in the causal past of  $x$  influences  $W(x, x)$  with no exponential suppression. No reduction of the response kernel to a purely spatial operator on  $\Sigma$  is constructed, so the exponential-decay bound on  $\mathcal{K}_{\text{nl}}$  is not established as stated. This is the principal gap.

*This gap is structural, not presentational* (investigated in `explorations/2026-06-16-FPE4-M3-stru`). The non-local kernel is the image of a source under a Green operator of the *normally hyperbolic* operator  $(\square_g + m^2 + \xi R)$ , whose support is the causal past  $J^-(x)$  (for the bare Green operator; the renormalized-kernel support has not been established to be smaller); this is an operator-support property, independent of the field mass (the compact- $\Sigma$  spectral gap  $\lambda_1(\Sigma)$  and the conformal shift  $\xi R$  control the *spatial* elliptic operator, which is the wrong operator, so the gap is not cured in the massless limit either). Reducing this object to an  $H^s(\Sigma) \rightarrow H^{s-2}(\Sigma)$  operator presupposes a canonical decomposition of the Lorentzian Green operator into time evolution off a chosen  $\Sigma$  — i.e. a global time function or homogeneous slicing. Assumptions (A1)–(A6) supply neither (this is the same unjustified background structure flagged as Gap (M8): a genuine spatial reduction of the kernel would also exhibit the chosen  $\Sigma$  as pure gauge, and conversely). The only known route to a contraction for the Lorentzian object — Meda–Pinamonti–Siemssen on FLRW [16, 14] — contracts in a norm over an interval of cosmological *time* on the Hubble function, exploiting homogeneous slicing, and is not a single-Cauchy-surface  $H^s$  bound. **Consequently the perturbative contraction of this section is permanently Sketch at the stated generality:** a repair requires geometric input beyond (A1)–(A6) — a controlled global foliation — and is then foliation-restricted, yielding existence “Rigorous on a [homogeneous / controlled] slicing,” local in time, rather than the general statement of Theorem 2. This does not refute existence of the fixed point, which holds exactly on de Sitter (Section 3) and on FLRW [16]; it obstructs only the single-slice spatial-resolvent *proof strategy* written here.

**Gap (M4): asserted coefficient scaling in Step 3.** The bound  $\|a_\gamma\|_{L^\infty} \leq C_{\text{loc}} \cdot \hbar m^2 / (4\pi)^2$  is asserted, not derived. It fails for the mass-independent anomaly counterterm responses: the variation of the  $R^2$ -,  $C^2$ -, and Euler-type counterterms in  $C_{\mu\nu}[g]$  carries coefficients  $\sim \hbar / (4\pi)^2$  times curvature, contributing local terms scaling as  $\hbar R$ , not  $\hbar m^2$ . These can be absorbed into  $C_\mathcal{K}$  only at the cost of a factor  $R/m^2$ , which makes the displayed  $m^2$  scaling — and hence the  $(m/M_P)^2$  headline of Theorem 2 — cosmetic whenever  $R \gtrsim m^2$ . A derivation of the coefficient bounds with explicit mass and curvature dependence is required. This is the same fork the (M1) diagnosis reaches from the operator-norm side: fixing the single Sobolev reference scale  $\mu^2 \sim R$  that makes the  $\kappa$  chain consistent forces the local-part scale to  $\hbar R / (4\pi)^2$ . (M1) and (M4) must therefore be repaired jointly.

*Remark on the massless case.* The binding obstruction (Gap M3) is mass-independent: the non-local response kernel is a Green operator of the Lorentzian hyperbolic operator  $(\square_g + m^2 + \xi R)$  with causal-past support  $J^-(x)$ , and no controlled reduction to a spatial  $H^s(\Sigma)$  operator exists within (A1)–(A6) regardless of whether  $m > 0$  or  $m = 0$ . The algebraic-decay and Hilbert–Schmidt arguments that appear in the massive Sketch are downstream of the spatial-operator assumption that Gap (M3) already identifies as illegitimate; they do not constitute an independent infrared obstruction. The massless case therefore fails for the same structural reason as the massive case, and remains open on the same terms.

*Remark on the Källén–Lehmann approach.* On Minkowski spacetime, the bound can alternatively be established via the Källén–Lehmann spectral representation of the stress-energy two-point function [5, 6]: the spectral gap above  $4m^2$  provides polynomial suppression at high momenta, and the Schur test translates this to an  $H^s \rightarrow H^{s-2}$  bound. This approach is rigorous on flat backgrounds but does not extend directly to general curved spacetimes, where no Källén–Lehmann representation for the response kernel is

available. The Hadamard decomposition above avoids this difficulty.

*Remark on related work.* Meda, Pinamonti, and Siemssen [16] established a Banach contraction for the semiclassical Einstein equation on FLRW spacetimes, using symmetry-reduced bounds specific to cosmological backgrounds. Meda and Pinamonti [17] showed that massive fields give polynomial decay of perturbations. The present Lemma addresses the general (non-symmetric) case via the Hadamard decomposition.

**Theorem 2** (Perturbative existence—massive fields — **Sketch**). *Let  $\bar{g}$  be a classical solution with sub-Planckian curvature  $R \ll \ell_P^{-2}$ , admitting a compact Cauchy surface, and let the matter content consist of massive fields with  $m > 0$ . Under the hypotheses of Lemma 1, the self-consistency map  $\mathcal{F}$  is a contraction in a neighborhood of  $\bar{g}$  in  $H^s$ , with Lipschitz constant*

$$\kappa = 8\pi G \cdot C_{\text{inv}} \cdot \sup_{g \in K_\rho} \|\mathcal{K}^{\text{red}}[g]\|_{H^s \rightarrow H^{s-2}} \leq C \cdot \frac{1}{2\pi} \left( \frac{m}{M_P} \right)^2 \ell_P^2 R \ll 1 \quad (8)$$

where  $C_{\text{inv}}$  is the norm of the inverse linearized Einstein operator from (A4) and  $C$  is a dimensionless constant. By the Banach fixed-point theorem [11], there exists a unique self-consistent solution  $g^*$  in a neighborhood of  $\bar{g}$ , and the iteration  $g_{n+1} = \mathcal{F}(g_n)$  converges exponentially:  $\|g_n - g^*\|_{H^s} \leq \kappa^n \|g_0 - g^*\|_{H^s}$ .

*Proof.* The Lipschitz constant of  $\mathcal{F}$  is

$$\kappa = 8\pi G \|(G_{\text{lin}})^{-1}\|_{H^{s-2} \rightarrow H^s} \cdot \sup_{g \in K_\rho} \|\mathcal{K}^{\text{red}}[g]\|_{H^s \rightarrow H^{s-2}}. \quad (9)$$

By assumption (A4), the inverse linearized Einstein operator satisfies  $\|(G_{\text{lin}})^{-1}\|_{H^{s-2} \rightarrow H^s} \leq C_{\text{inv}}$ , with  $C_{\text{inv}} \sim 1/R$ , since the harmonic gauge condition (A4) renders the linearized Einstein operator elliptic with principal symbol determined by the background curvature scale  $R$ ; standard elliptic regularity estimates then give an inverse operator norm scaling as  $1/R$ . By Lemma 1, the response kernel norm is bounded by  $C_K \cdot \hbar m^2 / (4\pi)^2$ . Combining:

$$\kappa \leq 8\pi G \cdot \frac{C_{\text{inv}} C_K \hbar m^2}{(4\pi)^2} \sim \frac{\hbar G m^2}{2\pi} \cdot \frac{1}{R} = \frac{1}{2\pi} \left( \frac{m}{M_P} \right)^2 \ell_P^2 R. \quad (10)$$

For sub-Planckian curvature ( $R \ll \ell_P^{-2}$ , i.e.,  $\ell_P^2 R \ll 1$ ) and any known particle mass ( $m \ll M_P$ ), we have  $\kappa \ll 1$ . The Banach fixed-point theorem [11] then guarantees a unique fixed point in the closed ball around  $\bar{g}$  of radius  $\rho/(1 - \kappa)$ , with geometric convergence.  $\square$

**Gap (M1): dimensional inconsistency in the  $\kappa$  chain.** The proof asserts  $C_{\text{inv}} \sim 1/R$  (dimension  $[\text{length}]^2$ , consistent with an  $H^{s-2} \rightarrow H^s$  operator norm), and then writes  $\kappa \sim (\hbar G m^2 / 2\pi) \cdot (1/R) = (1/2\pi)(m/M_P)^2 \ell_P^2 R$ . The two sides of this equality are different functions of  $R$  with different dimensions: with  $\hbar G = \ell_P^2$ , the left side is  $\ell_P^2 m^2 / (2\pi R)$  (dimension  $[\text{length}]^2$ ) while the right side is dimensionless; they coincide only if  $1/R = \ell_P^2 R$ , i.e. the step silently sets  $R = \ell_P^{-2}$ . Followed literally, the chain gives  $\kappa \propto 1/R$ , which *diverges* in the flat-space limit  $R \rightarrow 0$  — contraction would fail near Minkowski space, which is physically absurd and contradicts the displayed formula  $\kappa \propto R$ . Which  $R$ -scaling (if either) is correct is open; until the chain is repaired, the

quantitative form of eq. (8), and every claim downstream of it (notably Section 7), is unsupported.

*Resolution (Sketch).* The inconsistency is a hidden-reference-scale artifact. A dimensionally homogeneous Sobolev norm carries a reference mass  $\mu$ ,  $\|u\|_{H^t}^2 = \langle u, (\mu^2 - \Delta)^t u \rangle$ , so an operator norm  $\|A\|_{H^a \rightarrow H^b}$  carries an explicit factor  $\mu^{a-b}$  times the intrinsic dimension of  $A$ . Then  $C_{\text{inv}} = \|(G_{\text{lin}})^{-1}\|_{H^{s-2} \rightarrow H^s}$  carries  $\mu^{-2}$  (its “ $1/R$ ” is really  $1/(\text{spectral gap of } G_{\text{lin}})$ , equal to  $1/R$  only when  $\mu^2 \sim R$ ), while  $\|\mathcal{K}^{\text{red}}\|_{H^s \rightarrow H^{s-2}}$  carries  $\mu^{+2}$  (the asserted  $\hbar m^2/(4\pi)^2$  is licensed only when  $\mu^2 \sim m^2$ ). The displayed chain uses *two different* implicit scales —  $\mu^2 = R$  in  $C_{\text{inv}}$  and  $\mu^2 = m^2$  in the kernel — which is exactly why the unbalanced factor  $\ell_P^2 R$  appears. A *single* explicit  $\mu$  in both factors removes the inconsistency, at the cost of a physical choice that collides with Gap (M4):

- $\mu^2 \sim m^2$  (the field’s Compton scale):  $\kappa \sim (m/M_P)^2$ , *independent of  $R$* , with a graceful flat-space limit ( $\kappa$  unaffected as  $R \rightarrow 0$ ; no lower bound on  $R$  is needed). *This is schematic:* the Sobolev-index factors  $\mu^{-2}$  (from  $C_{\text{inv}}$ ) and  $\mu^{+2}$  (from  $\|\mathcal{K}^{\text{red}}\|$ ) cancel, but the surviving net physical scale depends on the intrinsic dimensions of the operator product, which requires the joint (M1)+(M4) treatment to make precise. The corrected Sketch-level bound below is the precise statement.
- $\mu^2 \sim R$ :  $\kappa \sim \ell_P^2 R$  (also  $\rightarrow 0$  as  $R \rightarrow 0$ ), but consistency then forces the kernel scale to  $\hbar R/(4\pi)^2$ , not  $\hbar m^2/(4\pi)^2$  — precisely Gap (M4). The  $(m/M_P)^2$  headline does not survive this reading.

The corrected Sketch-level statement is  $\kappa \leq C (m/M_P)^2 f(\mu^2/m^2, \ell_P^2 R)$  with  $f \rightarrow \text{const}$  for  $\mu^2 \sim m^2$  and  $f \sim R/m^2$  for  $\mu^2 \sim R$ . The theorem requires an explicit single  $\mu$ , fixed jointly with (M4) — not a lower bound on  $R$ . The displayed  $\kappa = (1/2\pi)(m/M_P)^2 \ell_P^2 R$  is an artifact of mixing the two scales and then setting  $R = \ell_P^{-2}$ ; it is not a derivation. (This confirms the Section 7 withdrawal from a second direction: on the defensible  $\mu^2 \sim m^2$  reading  $\kappa$  is  $R$ -independent, so no breakdown curvature is derived at all.)

**Gap (M5): missing self-mapping step.** The invocation of the Banach theorem on “the closed ball around  $\bar{g}$  of radius  $\rho/(1 - \kappa)$ ” requires, in addition to the contraction estimate, the displacement bound  $\|\mathcal{F}(\bar{g}) - \bar{g}\|_{H^s} \leq (1 - \kappa)\rho$  — i.e. that the quantum correction evaluated *at the background* is small in  $H^s$ . This is never established; it is essentially an (A3)-type bound at  $\bar{g}$ . Contraction alone does not yield existence without it. This is a genuine technical gap, subordinate to (M3): it would still have to be discharged for any foliation-restricted repair (Section 8), in the time-weighted norm that repair uses rather than in  $H^s(\Sigma)$ .

For all known particles,  $\kappa_{\text{mass}} \equiv (m/M_P)^2 \ll 1$ :

| Particle  | Mass (kg)             | $\kappa_{\text{mass}} = (m/M_P)^2$ |
|-----------|-----------------------|------------------------------------|
| Electron  | $9.1 \times 10^{-31}$ | $\sim 10^{-45}$                    |
| Proton    | $1.7 \times 10^{-27}$ | $\sim 10^{-39}$                    |
| Top quark | $3.1 \times 10^{-25}$ | $\sim 10^{-34}$                    |



## 5 Non-perturbative existence via Schauder

### 5.1 Setup

Let  $(\mathcal{M}, \bar{g})$  be a globally hyperbolic spacetime with *compact* Cauchy surface  $\Sigma$ . Fix  $s > n/2 + 2 = 4$  and define the candidate fixed-point set:

$$K_\rho \equiv \{h \in H^s(\Sigma) : \|h - \bar{g}\|_{H^s} \leq \rho\} \cap \mathcal{G}_{\text{harm}} \cap \mathcal{L} \quad (11)$$

where  $\mathcal{G}_{\text{harm}}$  is the closed affine subspace of metrics in linearized harmonic gauge and  $\mathcal{L}$  is the open set of Lorentzian-signature metrics. The required convexity and compactness of  $K_\rho$  are established in the proof of Theorem 3 below: for  $\rho$  below a threshold set by the Sobolev embedding constant,  $\bar{B}_\rho(\bar{g}) \subset \mathcal{L}$  and the intersection with  $\mathcal{L}$  is vacuous, reducing  $K_\rho$  to the intersection of a closed convex ball with a closed affine subspace — hence closed and convex in  $H^s$ , compact in  $H^{s'}$  for  $s' < s$  by Rellich–Kondrachov.

### 5.2 Assumptions

- (A1) *Hadamard state existence.* For each  $g \in K_\rho$ , a Hadamard state  $|\Psi_g\rangle$  exists and  $\langle \hat{T}_{\mu\nu} \rangle_{g, \text{ren}}$  is well-defined. **[Established:** Radzikowski [7], Fulling–Sweeny–Wald [8].]
- (A2) *Continuous metric dependence.* The map  $g \mapsto \langle \hat{T}_{\mu\nu} \rangle_{g, \text{ren}}$  is continuous  $H^s \rightarrow H^{s-2}$ . **[Established:** Hollands–Wald [9].]
- (A3) *Uniform stress-energy bound.* There exists  $B(\rho) < \infty$  with  $\|\langle \hat{T}_{\mu\nu} \rangle_{g, \text{ren}}\|_{H^{s-2}} \leq B(\rho)$  for all  $g \in K_\rho$ . **[Conjecture.** Established for FRW spacetimes [14]; open in general.]
- (A4) *Linearized Einstein well-posedness.* The linearized Einstein equation in harmonic gauge has a unique solution  $h \in H^s$  with  $\|h\|_{H^s} \leq C_{\text{inv}} \|S\|_{H^{s-2}}$ . **[Established:** Choquet-Bruhat [10].]
- (A5) *Order reduction.* Fourth-derivative terms from the trace anomaly can be perturbatively eliminated. **[Established:** Parker–Simon [13].]
- (A6) *Uniform contraction bound.* (Included for completeness; used only in the Banach uniqueness argument of Section 4, not in the Schauder existence argument below.)  $\kappa_{\text{sup}} \equiv 8\pi G C_{\text{inv}} \cdot \sup_{g \in K_\rho} \|\delta \langle \hat{T}_{\mu\nu} \rangle_{\text{ren}} / \delta g\|_{\text{op}} < 1$ . **[Sketch for massive fields, with a structural obstruction.** Lemma 1 sketches the operator-norm bound. The displayed target  $\kappa \sim (m/M_P)^2 \ell_P^2 R$  is not derived as written (Gap (M1): mixing two Sobolev reference scales; the consistent single-scale form is  $\kappa \sim (m/M_P)^2$ ,  $R$ -independent). The governing gap (M3) is *structural*: the response kernel is a Lorentzian hyperbolic Green operator with causal-past support, and its reduction to a spatial operator on  $\Sigma$  requires a controlled global foliation beyond (A1)–(A6). The bound holds only on such a (foliation-restricted) slicing; see Gaps (M1), (M3), (M4), (M5) in Section 4. Open for massless fields (the same (M3) gap binds).]

### 5.3 Theorem

**Theorem 3** (Non-perturbative existence (conditional on (A3))). *Let  $(\mathcal{M}, \bar{g})$  be globally hyperbolic with compact Cauchy surface  $\Sigma$ , and  $\bar{g}$  a classical solution in linearized*

harmonic gauge. Under assumptions (A1)–(A5) and the semiclassical hypothesis  $\hbar < \hbar_{\max}$  (15), the self-consistency map  $\mathcal{F}$  has at least one fixed point  $g^* \in K_{\rho^*}$  in the  $H^{s'}$  topology for any  $s' < s$ , where  $\rho^* = C_{\text{inv}} B_0 \hbar$  (14). Assumption (A3) enters through the uniform bound used in Step 2.

*Proof. (Rigorous, conditional on (A3).)* We verify the hypotheses of the Schauder fixed-point theorem [12].

*Step 1:  $K_\rho$  is convex, closed, and compact (for  $\rho$  below a threshold).* Since  $s > n/2 + 2 = 4$  and  $\Sigma$  is compact, the Sobolev embedding  $H^s(\Sigma) \hookrightarrow C^2(\Sigma)$  holds with embedding constant  $C_{\text{emb}}$ :  $\|h\|_{C^2} \leq C_{\text{emb}} \|h\|_{H^s}$  for all  $h \in H^s(\Sigma)$ . The background  $\bar{g}$  is Lorentzian, so its eigenvalues  $\lambda_i(\bar{g}(x))$  satisfy  $\lambda_{\min} \equiv \min_{x \in \Sigma} \min_i |\lambda_i(\bar{g}(x))| > 0$ . For any  $g$  with  $\|g - \bar{g}\|_{H^s} \leq \rho$ , the  $C^0$ -distance satisfies  $\|g - \bar{g}\|_{C^0} \leq C_{\text{emb}} \rho$ , and eigenvalue perturbation gives  $|\lambda_i(g(x)) - \lambda_i(\bar{g}(x))| \leq C_{\text{emb}} \rho$  pointwise. Choose the Lorentzian-signature threshold:

$$\rho_{\mathcal{L}} \equiv \frac{\lambda_{\min}}{C_{\text{emb}}}. \quad (12)$$

For  $\rho < \rho_{\mathcal{L}}$ , every  $g$  in the closed  $H^s$  ball  $\bar{B}_\rho(\bar{g}) \equiv \{h \in H^s(\Sigma) : \|h - \bar{g}\|_{H^s} \leq \rho\}$  has signature identical to  $\bar{g}$  at every point of  $\Sigma$ ; in particular it is Lorentzian. Thus the intersection with  $\mathcal{L}$  is vacuous ( $\bar{B}_\rho(\bar{g}) \subset \mathcal{L}$ ), and

$$K_\rho = \bar{B}_\rho(\bar{g}) \cap \mathcal{G}_{\text{harm}} \quad \text{for } \rho < \rho_{\mathcal{L}}.$$

The closed ball  $\bar{B}_\rho(\bar{g})$  is closed and convex in  $H^s$ ;  $\mathcal{G}_{\text{harm}}$  is a closed affine subspace (the linearized harmonic-gauge condition  $\nabla^\mu(h_{\mu\nu} - \frac{1}{2}\bar{g}_{\mu\nu}\text{tr } h) = 0$  is a closed linear constraint). Hence  $K_\rho$  is closed and convex in  $H^s$ . By the Rellich–Kondrachov theorem on compact  $\Sigma$ , the embedding  $H^s(\Sigma) \hookrightarrow H^{s'}(\Sigma)$  is compact for  $s' < s$ , so the closed bounded set  $K_\rho$  is compact in the  $H^{s'}$  topology.

*Step 2:  $\mathcal{F}$  maps  $K_\rho$  into  $K_\rho$  (semiclassical  $\hbar$ -smallness regime).* Given  $g \in K_\rho$ : (A1) provides a Hadamard state with well-defined renormalized stress-energy; (A5) reduces the fourth-order system to second order; (A4) yields  $\mathcal{F}(g) \in H^s$  with  $\|\mathcal{F}(g) - \bar{g}\|_{H^s} \leq C_{\text{inv}} \|S_{\mu\nu}[g]\|_{H^{s-2}} \leq C_{\text{inv}} B(\rho)$  by (A3).

*$\rho$ -selection.* The renormalized stress-energy carries an overall factor of  $\hbar$  from the one-loop structure of  $\langle \hat{T}_{\mu\nu} \rangle_{\text{ren}}$ :

$$B(\rho) \leq B_0 \hbar \quad \text{for a constant } B_0 \text{ depending on } (\bar{g}, m, \xi, s) \text{ but not on } \rho, \quad (13)$$

where the  $\rho$ -independence holds because the leading-order stress-energy is determined by the background geometry (the state-dependence on  $g$  contributes at  $O(\hbar\rho)$  — a subleading correction that is absorbed into  $B_0$  for  $\rho \leq 1$ ). The self-mapping condition  $C_{\text{inv}} B(\rho) \leq \rho$  is therefore satisfied by choosing

$$\rho^* = C_{\text{inv}} B_0 \hbar. \quad (14)$$

In the semiclassical regime  $\hbar \ll \lambda_{\min}/(C_{\text{emb}} C_{\text{inv}} B_0)$ , this gives  $\rho^* < \rho_{\mathcal{L}}$ , so Step 1 applies. The condition on  $\hbar$  is stated explicitly:

$$\hbar < \hbar_{\max} \equiv \frac{\lambda_{\min}}{C_{\text{emb}} C_{\text{inv}} B_0}. \quad (15)$$

This is the *semiclassical hypothesis*: the quantum stress-energy must be small enough that the geometry is not driven away from the Lorentzian signature of the background.

*Harmonic-gauge preservation.* The output  $\mathcal{F}(g)$  lies in  $\mathcal{G}_{\text{harm}}$  by construction: (A4) solves the linearized Einstein equation *in* harmonic gauge — the gauge condition is built into the solution operator, not an ex post constraint on the output. Explicitly, (A4) inverts the reduced (gauge-fixed) Einstein operator, whose solution automatically satisfies the linearized de Donder condition  $\nabla^\mu(\mathcal{F}(g)_{\mu\nu} - \frac{1}{2}\bar{g}_{\mu\nu}\text{tr}\mathcal{F}(g)) = 0$ ; the contracted Bianchi identity  $\nabla^\mu G_{\mu\nu} = 0$  together with  $\nabla^\mu S_{\mu\nu} = 0$  (the latter following from the Ward identity for  $\langle\hat{T}_{\mu\nu}\rangle_{\text{ren}}$  evaluated at the solution  $g$ ) ensures the gauge condition propagates. Therefore  $\mathcal{F}(g) \in \mathcal{G}_{\text{harm}}$ , and combined with the norm bound above,  $\mathcal{F}(g) \in K_{\rho^*}$ .

*Step 3:  $\mathcal{F}$  is continuous.* By (A2),  $g \mapsto S_{\mu\nu}[g]$  is continuous in  $H^{s-2}$ ; by (A4), the Einstein solve is continuous; composing,  $\mathcal{F} : K_{\rho^*} \rightarrow K_{\rho^*}$  is continuous in  $H^{s'}$ .

By the Schauder theorem [12],  $\mathcal{F}$  has a fixed point in  $K_{\rho^*}$ .  $\square$

*Remark.* Assumption (A6) is not used in the Schauder argument (which requires only continuity, not contractivity). It would be needed for a Banach fixed-point argument guaranteeing uniqueness and constructive convergence — an argument that is itself presently a Sketch (Section 4).

## 6 Effective initial-value formulation

Theorem 3 establishes global existence conditionally on (A3). Practical physics requires a Cauchy formulation: given data on a spacelike hypersurface, what is the evolution? The global fixed-point condition does not directly reduce to a Cauchy problem because the Hadamard state is defined by a global microlocal condition. We derive an effective formulation valid order-by-order in  $\kappa$ .

At order  $\kappa^0$ , the evolution is classical GR with given matter initial data.

At order  $\kappa^1$ , the semiclassical correction  $\langle\hat{T}_{\mu\nu}\rangle^{(1)}$  is computed on the order- $\kappa^0$  background geometry. By Parker–Simon [13] order reduction, the resulting system is second-order in time, with initial data for  $g_{\mu\nu}$  and  $\partial_t g_{\mu\nu}$  on  $\Sigma$  supplemented by the adiabatic vacuum state at each order.

The order-reduced system at each order  $\kappa^n$  is a second-order hyperbolic PDE, well-posed by Choquet-Bruhat [10]. The perturbative expansion is valid while  $\kappa \ll 1$ ; where that regime ends is unresolved pending the corrected contraction estimate (Section 7).

**Open problem.** Non-perturbative well-posedness of the order-reduced semiclassical Einstein equation on general globally hyperbolic spacetimes without symmetry assumptions has been conjectured by Juárez-Aubry et al. [15] and established for cosmological spacetimes [14]. The general case remains open.

## 7 Validity boundary: a withdrawn claim

An earlier revision of this paper claimed that the contraction constant (8) satisfies  $\kappa \rightarrow 1$  when  $R \sim \ell_P^{-2}$ , so that the Planck scale emerges as the natural validity boundary of the framework — derived from the convergence requirement  $\kappa < 1$  rather than imposed by hand. **This claim is withdrawn.** It is contradicted by the paper’s own formulas on either reading of the  $\kappa$  chain (Gap (M1), Section 4):

- Taking the displayed formula  $\kappa = (1/2\pi)(m/M_P)^2\ell_P^2 R$  at face value, at Planck curvature  $R = \ell_P^{-2}$  one finds  $\kappa = (m/M_P)^2/2\pi$ , which by the table in Section 4 is

at most  $\sim 10^{-34}$  for known particles — nowhere near 1. On this reading,  $\kappa \rightarrow 1$  only at curvatures  $\sim (M_P/m)^2$  *above* the Planck scale.

- Following the derivation chain literally ( $C_{\text{inv}} \sim 1/R$ ), one finds  $\kappa \propto 1/R$ , which reaches 1 at curvature  $R \sim m^2$  — the Compton scale of the field, far *below* the Planck scale — and diverges in the flat-space limit.

Neither reading places the breakdown at the Planck scale, and the two disagree with each other. Until the contraction estimate is repaired (Section 8), the framework’s validity boundary must be regarded as underived. The qualitative expectation that semiclassical gravity fails at Planckian curvature remains physically plausible, but it is not a result of this paper.

The framework also fails for observables that require summing over spacetime topologies (e.g., the Page curve of black hole evaporation). Even when  $\kappa \ll 1$ —so the perturbative expansion is valid—certain correlators receive  $\mathcal{O}(1)$  contributions from topology-changing saddles absent from the fixed-topology description. This is a limitation distinct from the (unresolved) convergence-boundary question above, and is not self-diagnosable within the framework.

## 8 Open problems

1. **The contraction estimate (Gaps M1, M3, M4, M5): a foliation-restricted repair is the honest path.** The Banach argument of Section 4 is permanently a Sketch at the stated generality. The gaps are not on equal footing (analysis: `explorations/2026-06-16-FPE4-M3-structural-obstruction.md`). (*M3*) is *structural*: the non-local response kernel is a Green operator of the Lorentzian hyperbolic operator, with causal-past support, and no reduction to a spatial operator on a single Cauchy surface  $\Sigma$  exists within (A1)–(A6) (sharing its root with (M8) — two faces of one obstruction: a genuine spatial reduction would simultaneously close M8’s well-definedness gap; the obstruction is mass-independent, so it is not cured in the massless limit). The only known repair replaces the single-slice  $H^s(\Sigma)$  bound by a time-weighted energy (Volterra) estimate on a finite causal slab, following Meda–Pinamonti–Siemssen [16, 14]; this requires a controlled global foliation — geometric input beyond (A1)–(A6) — and yields existence “Rigorous on a [homogeneous / controlled] slicing,” local in time, not the general statement of Theorem 2. (*M1*) is a *hidden-reference-scale artifact* (diagnosed in Section 4; corrected form schematic, net scale tied to the joint (M4) repair): with a single explicit Sobolev scale  $\mu$  the Sobolev-index mismatch is removed, giving  $\kappa \sim (m/M_P)^2$  ( $R$ -independent, schematic) for  $\mu^2 \sim m^2$ , or  $\kappa \sim \ell_P^2 R$  for  $\mu^2 \sim R$  — the latter forcing the local scale  $\hbar R/(4\pi)^2$ , i.e. (*M4*), so (M1) and (M4) are one joint repair. (*M5*) (the self-mapping/displacement bound) remains a genuine technical gap that any repair must still discharge, in the norm that repair uses. Restoring a Rigorous label at the general stated scope is therefore not available; a foliation-restricted re-promotion is.
2. **Well-definedness of the map  $\mathcal{F}$  (M8, M9).** Two structural gaps predate the demotion and remain open. (*M8*) Sections 4–5 work with fields in  $H^s(\Sigma)$  on a single Cauchy surface, while  $\mathcal{F}$  is defined on spacetime geometries; the equivalence

between a spacetime fixed point and a fixed point of data on a chosen  $\Sigma$  is never constructed (the Hadamard condition is global), and the role of the chosen  $\Sigma$  must be shown to be pure gauge. (M9) The single-valuedness of  $\mathcal{F}$  rests on the assertion that the Hadamard state on the corrected geometry is “the unique perturbation” of the background state; Hadamard states are not unique, and the perturbation prescription that would make this precise is assumed, not constructed. Relatedly, the “Established” tags on (A1) and (A2) are generous: Hollands–Wald [9] establishes local/analytic metric dependence of Wick polynomials in the algebraic sense, not an  $H^s \rightarrow H^{s-2}$  continuity statement, and the deformation existence argument behind (A1) should be attributed and verified precisely.

3. **Extension to massless fields.** The binding obstruction for the massless extension is Gap (M3), not an independent infrared failure. Gap (M3) is mass-independent: the non-local response kernel is a Lorentzian Green operator with causal-past support regardless of whether  $m > 0$ , and the algebraic-decay and Hilbert–Schmidt arguments that appear in the massive Sketch are downstream of the spatial-operator assumption Gap (M3) already identifies as illegitimate. Closing the massless extension therefore requires discharging (M3) first; any remaining IR issues (e.g. algebraic kernel decay when  $m = 0$ ) would be a secondary obstacle, not the present bottleneck.
4. **Uniform stress-energy bound (A3).** The Pinamonti–Siemssen result [14] establishes this for FRW spacetimes. Extension to general globally hyperbolic spacetimes with compact Cauchy surfaces requires controlling the state-dependent part of  $\langle \hat{T}_{\mu\nu} \rangle_{\text{ren}}$  uniformly over  $K_\rho$ —an open problem.
5. **Relaxing the compact Cauchy surface assumption.** Both the Banach argument (via Step 6 of Lemma 1 and the Hilbert–Schmidt bound) and the Schauder argument require compactness of  $K_\rho$ , ensured by Rellich–Kondrachov on compact  $\Sigma$ . Neither result therefore covers Minkowski space or asymptotically flat backgrounds. Extension requires weighted Sobolev spaces and a different compactness argument.
6. **Non-perturbative IVP well-posedness.** See Section 6.

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